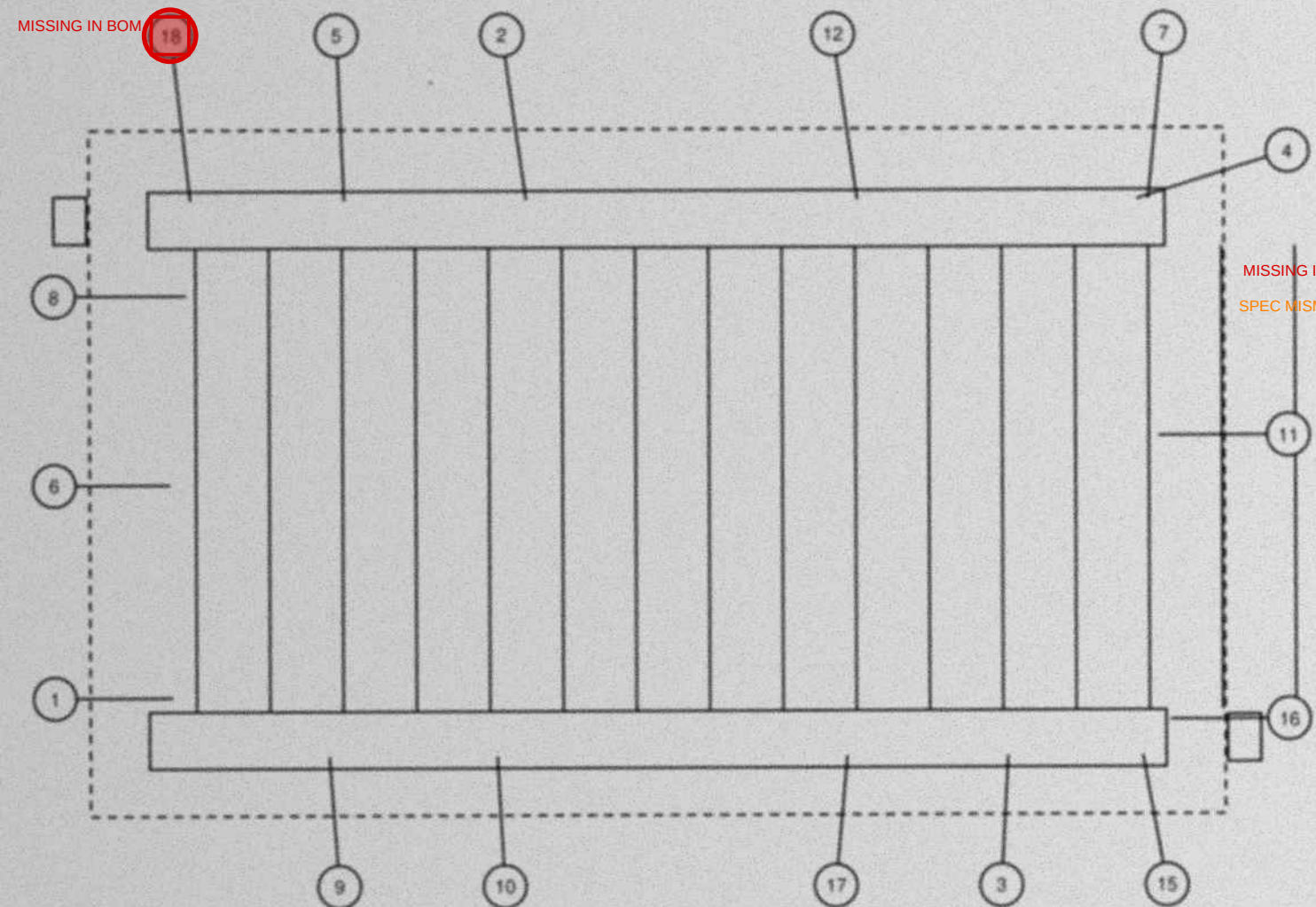


BOM CHECK (automated - verify before acting)

- red = on drawing, missing in BOM
- orange = qty / spec / revision mismatch
- yellow = confirm match / extra / duplicate
- blue = suggestion (implied item, rule-based)

RUN RELIABILITY: LOW - DO NOT RELY ON THIS RUN



Item	Description	Material	Qty	Part No	Remarks
1	INLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1	EC-01-01	
2	OUTLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1	EC-01-02	
3	ECONOMISER TUBE 38.1 OD X 3.6 THK	SA 210 GR.A1	138	EC-01-03	
4	RETURN BEND 38.1 OD	SA 210 GR.A1	30	EC-01-04	
5	TUBE SUPPORT PLATE 12 THK	IS 2062 GR.B	14	EC-01-05	
6	INLET FLANGE 200 NB CL.150	SA 105	1	EC-01-06	
7	OUTLET FLANGE 200 NB CL.150	SA 105	1	EC-01-07	
8	SAFETY VALVE 50 NB	SA 216 WCB	1	EC-01-08	
9	DRAIN VALVE 25 NB	SA 105	2	EC-01-09	
10	VENT VALVE 15 NB	SA 105	1	EC-01-10	
11	THERMOWELL 1/2 NPT	SS 316	2	EC-01-11	
12	PRESSURE TAPPING 1/2 NB	SA 105	2	EC-01-12	
13	GASKET 200 NB CL.150	SS 304 + GRAPHITE	2	EC-01-13	
14	STUD BOLT M20 X 90 WITH 2 NUTS	SA 193 B7	16	EC-01-14	
15	LIFTING LUG 16 THK	IS 2062 GR.B	4	EC-01-15	
16	EXPANSION JOINT 600 X 600	FABRIC	1	EC-01-16	
17	NAME PLATE	SS 304	1	EC-01-17	
18	INSPECTION DOOR 300 X 300	IS 2062	2	EC-01-18	

NOTES:

- ALL DIMENSIONS ARE IN MM.
- ALL WELDING AS PER IBR / ASME SEC I. <- BOM LACKS: Welding consumables
- INSULATE WITH 100 THK MINERAL WOOL.

ECONOMISER ASSEMBLY

DRG NO : EC-01-GA-001
DRG REV : D
SCALE : NTS SIZE : A3
SAMPLE (SYNTHETIC) DRAWING

BOM CHECK SUMMARY

Drawing : sample_01_drawing_scan.pdf (Rev ?)

BOM : sample_01_bom.xlsx (Rev D)

Input : scanned (OCR)

RUN RELIABILITY : LOW

- only 2 parts-table rows were read
- only 5% of BOM lines could be linked to a drawing item (wording/extraction problem?)
- low OCR confidence (52/100)

Drawing items checked : 2

BOM lines read : 20

BOM completeness : 50.0% (drawing items that have a BOM line)

Findings: extra 18, implied 1, omission 2, spec 1

[RED] Item 1 'INLET 219.1 00 X 8.0 THK GA' (qty 1) is on the drawing but has no BOM line [low OCR confidence - verify on sheet]

[RED] Balloon 18 on the drawing is in neither the parts table nor the BOM

[ORANGE] Item 2: specification mismatch - dimension: drawing ['0', '6'] vs BOM ['8']

[YELLOW] BOM row 6 'Header, inlet, 219.1 x 8.0 mm' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 8 'Tube, economiser, 38.1 x 3.6 mm' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 9 'U-bend, 38.1 OD' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 10 'Tube support plate 12 thk' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 11 'Flange, inlet, WNRF, 250 NB, 150#' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 12 'Flange, outlet, WNRF, 200 NB, 150#' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 13 'Safety relief valve, 50 NB' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 14 'Drain valve 25 NB' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 15 'Globe valve, vent, 15 NB' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 16 'Pressure gauge tapping, 1/2 NB' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 17 'Spiral wound gasket 200 NB Class 150' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 18 'Stud bolt M20 x 90 with 2 nuts' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 19 'Lug, lifting, 16 mm' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 20 'Expansion joint 600 x 600' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 21 'Name plate' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 22 'Inspection door, 300 x 300' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 24 'Gate valve, drain, 25 NB' has no matching item on the drawing (extra, or from another revision)

[YELLOW] BOM row 25 'Hand rail, 42 NB' has no matching item on the drawing (extra, or from another revision)

[BLUE] Suggestion (Welding consumables): the drawing notes imply it, but no matching line was found in the BOM

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.